

Shor's Algorithm, Read in the Right Basis

Phase kickback, one circuit with two descriptions, and the arithmetic that builds it

Where we are going

The whole story, once, so you know the shape

Multiplication by a mod N is a permutation, hence a unitary U_a ; on the orbit of 1 it is a **cyclic shift of length r** — the period we want. Shifts are diagonal in the Fourier basis, so U_a 's eigenvalues are $e^{2\pi is/r}$: **the period is an angle**. A controlled gate turns that angle into a measurable relative phase (*kickback*); the input $|1\rangle$ is secretly the sum of all r eigenvectors; an inverse Fourier transform converts the angle into a position you can read.

Three acts:

1. The five facts of quantum mechanics you actually need. (0–13 min)
2. The mechanism: bases, kickback, the spectrum of U_a , interference. (13–46 min)
3. The engineering: reversible arithmetic, the bill, the readout. (46–60 min)

I. The quantum mechanics you need

Why anyone thought of this: 1980–81

Two people, independently, noticed the same thing.

Yuri Manin (1980) — mathematician.

Describing n quantum degrees of freedom takes $\sim 2^n$ numbers. That is a *big state space to compute in*.

Richard Feynman (1981) — physicist.

Same fact, opposite reading: a classical machine *cannot* simulate quantum physics. Fight fire with fire.

Hold on to this

That 2^n is not a metaphor. In five minutes it will be a postulate — and it is the only place the exponential comes from.

First: what is a quantum state?

Quantum mechanics does not in general give a property a definite value. Name a measurable property, list the possible outcomes, and the state — written as a **wavefunction** ψ — assigns a complex **amplitude** to each outcome.

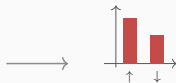
Born's rule: outcome v has probability $|\psi(v)|^2$, with $\sum_v |\psi(v)|^2 = 1$ — or, when the outcomes form a continuum, $|\psi|^2$ is a probability *density* and $\int |\psi|^2 = 1$.

Careful: ψ is the state written *in the basis of the property you named*. Pick a different property and the same state gives a different table. (That is moral M1, arriving early.)

position — a continuum of outcomes
 $|\psi(\mathbf{r})|^2$ is a probability *density*



spin- $\frac{1}{2}$ along an axis — two outcomes
 ψ is two complex numbers



Infinite outcome set (position) or finite (spin- $\frac{1}{2}$, polarisation, two *chosen* energy levels of an atom). **The gate model in this talk lives entirely in the finite case** — but the continuous case is not merely pedagogy: in continuous-variable photonics the quadratures $\hat{x}, \hat{p}$ of a light mode *are* the computational degree of freedom, and GKP codes store a qubit as a grid in x - p space. (backup slide)

Bits vs qubits

A **bit** is 0 or 1. Flip between them fast enough and you have a computer.

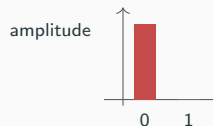
A **qubit** is the same — *plus* it can be a little bit of 0 and a little bit of 1.

Definition. A qubit is any physical system with two reliably distinguishable states (a spin, two energy levels, two polarisations). Its state is

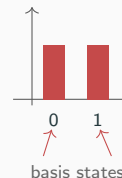
$$\alpha|0\rangle + \beta|1\rangle \in \mathbb{C}^2, \quad |\alpha|^2 + |\beta|^2 = 1.$$

α, β : *amplitudes*. $|0\rangle, |1\rangle$: *basis states*.

a definite bit



superposition



Exponentially many amplitudes

k qubits live in the tensor product

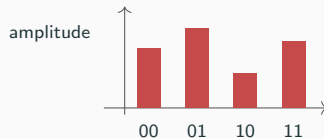
$$(\mathbb{C}^2)^{\otimes k} \cong \mathbb{C}^{2^k},$$

with one basis state per bit string, so

$$|\psi\rangle = \sum_x \psi_x |x\rangle, \quad \sum_x |\psi_x|^2 = 1.$$

Notation, fixed now:

- $|a\rangle \otimes |b\rangle \equiv |a\rangle|b\rangle \equiv |a, b\rangle$
- an integer names a bit string:
 $|x\rangle = |x_{k-1}\rangle \cdots |x_0\rangle, \quad x = \sum_i x_i 2^i$

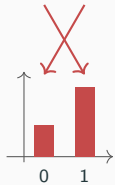


Adding one qubit **doubles** the number of bars.

This is Manin's and Feynman's exponential — and it is a *postulate*, not a theorem.

Quantum gates are maps over the amplitude table

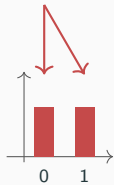
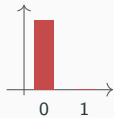
X: negation



$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



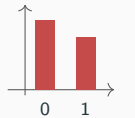
H: Hadamard



$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$



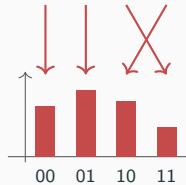
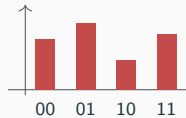
$P(\theta)$: phase



$$P(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$



CNOT



if the first bit is 1,
flip the second



Two things to notice about those gates

X and CNOT are permutations

Every matrix entry is 0 or 1. **No complex number appears.** Remember this: the entire arithmetic layer of Shor's algorithm is built from gates like these.

$P(\theta)$ moves no amplitude at all

It only multiplies the $|1\rangle$ bar by a point on the unit circle. **Measure right after it and nothing has changed.** A phase is invisible until an H turns it back into amplitude — so every useful phase sits *between* two Hadamard-like layers.

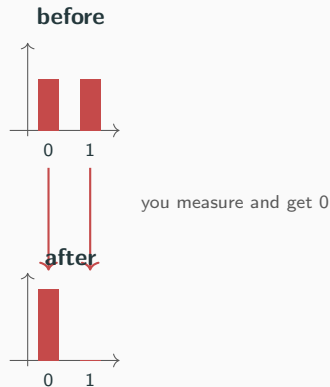
The rule that runs the whole talk. A gate must be **unitary**: $U^\dagger U = \mathbf{1}$, equivalently $\langle U\phi | U\psi \rangle = \langle \phi | \psi \rangle$. Lengths and angles preserved $\Rightarrow$ a circuit only ever **rotates** the state vector of $\mathbb{C}^{2^k}$. It moves amplitude around; it never creates it. Unitary also means *invertible* — every subroutine must be reversible. (We will pay for that in Act III.)

Measurement and “collapse”

Measuring returns x with probability $|\psi_x|^2$, and then you **condition** on what you saw and renormalise.

That is all. Physicists have a flair for the dramatic and say the state has **collapsed**.

Partial measurement: measure one register of many, and the surviving branch keeps *all* of its internal amplitudes and phases. Only *which-branch* information is destroyed.



Measurement is an interaction — and so is decoherence

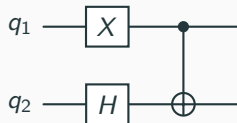
To learn where a particle is, you must interact with it: scatter a photon off it, or let it absorb one and see which energy level it lands in. **Information never arrives for free** — something must couple to the system and carry the record away.

That coupling correlates the system with the apparatus. Once a record exists, the alternatives it distinguishes **can no longer interfere**.

Decoherence is an uninvited measurement

A stray photon, a phonon, a fluctuating field — any of them can interact with a qubit and walk off with a record of which branch it was in. Not noise on top of a good answer: an *unrequested measurement performed by the surroundings*.

We will meet exactly this equation twice more: as leftover scratch data (Act II) and as the reason the machine must be error-corrected (Act III).



X q1;
H q2;
CNOT q1, q2;

How to read one:

- time flows left to right;
- each horizontal line is *one qubit*, for the whole computation;
- a vertical column is one instant;
- **nothing flows along the wires** — the picture is sheet music, not plumbing;
- • = control, $\oplus$ = target, a slashed line labelled n = a bundle of n qubits.

That is essentially all the quantum mechanics there is

A claim worth repeating (A. Albarghouthi, PLDI keynote)

Lock yourself in a room with those slides, no internet, enough food and patience, and you could derive essentially everything we know about quantum computing. *Maybe you need a couple more facts.*

The rest of this hour is that exercise, done for Shor's algorithm.

Here are the couple more facts.

Fact 1: interference — a theorem, not a postulate

If outcome x is reachable along several branches with amplitudes A_j :

$$\Pr[x] = \left| \sum_j A_j \right|^2 = \underbrace{\sum_j |A_j|^2}_{\text{classical}} + \underbrace{\sum_{j \neq j'} \overline{A_{j'}} A_j}_{\text{cross terms}}$$

Cross terms can be **negative**: paths cancel.

It follows from linearity plus the squaring in the measurement rule — nothing new was assumed.

This is the entire resource. Everything else in this talk is about *protecting these cross terms* long enough to use them.

Facts 2 and 3: two lemmas about unitaries

Lemma 1 — an eigenvalue of a unitary is an angle

If $U|u\rangle = \lambda|u\rangle$, $\langle u|u\rangle = 1$, then $1 = \langle u|U^\dagger U|u\rangle = |\lambda|^2$, so $\lambda = e^{2\pi i\varphi}$.

An eigenvalue carries no magnitude — only an angle. If a unitary is to tell us a number, that number *must* be encoded as a phase. Forced, not chosen.

Lemma 2 — distinct eigenvalues have orthogonal eigenvectors

$\langle u'|u\rangle = \langle u'|U^\dagger U|u\rangle = \overline{\lambda'}\lambda\langle u'|u\rangle$ and $\overline{\lambda'}\lambda \neq 1$.

Lemma 1 is why Shor's answer will be an angle. Lemma 2 is why one measurement will cleanly pick one branch out of r .

II. The mechanism

Three morals; watch them recur at every scale

M1 — operators are basis-free; matrices are not

Every unitary owns a basis — its eigenbasis — in which its matrix is **diagonal**: nothing mixes, each axis just spins by a phase. A unitary is a rotation; its eigenvectors are the axes.

M2 — a scalar has no address

$(\lambda|a\rangle) \otimes |b\rangle = |a\rangle \otimes (\lambda|b\rangle) = \lambda(|a\rangle \otimes |b\rangle)$. A phase belongs to a whole *branch*, not to either register. A phase on the whole state is unobservable; only *relative* phases are physical.

M3 — interference needs indistinguishable branches; collapse is conditioning

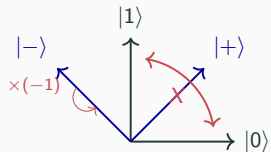
Two branches add amplitudes only if they agree on every register. Any leftover record multiplies their cross term by the overlap of the records.

M1 at size one: the same gate, two matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ in } \{|0\rangle, |1\rangle\}, \quad X = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ in } \{|+\rangle, |-\rangle\}, \quad X = HZH.$$

In the computational basis X is a **permutation** — it swaps the two axes, and no entry is complex: you cannot see a phase anywhere.

In its own eigenbasis the same operator **moves nothing**: it fixes $|+\rangle$ and multiplies $|-\rangle$ by $-1 = e^{i\pi}$.

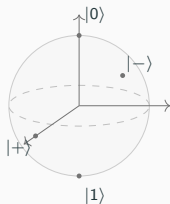


One rotation, two descriptions. H is the dictionary.

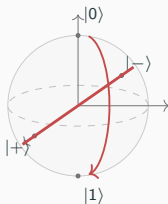
Theorem (spectral theorem)

Every unitary is $U = \sum_j e^{2\pi i \varphi_j} |u_j\rangle\langle u_j|$, i.e. $\text{diag}(e^{2\pi i \varphi_j})$ in its eigenbasis — because each eigenvector maps only to itself.

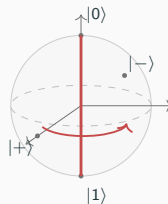
One qubit, honestly: the Bloch sphere



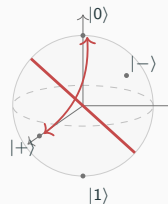
three axes
= three bases



X : half-turn about $\hat{x}$
 $|0\rangle \leftrightarrow |1\rangle$



Z : half-turn about $\hat{z}$
 $|+\rangle \leftrightarrow |-\rangle$



H : half-turn about $\frac{\hat{x}+\hat{z}}{\sqrt{2}}$
 $|0\rangle \leftrightarrow |+\rangle$

Every state $\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle$ is a point; **antipodal points are orthogonal kets**, so each axis carries one measurement basis. Every one-qubit unitary is a rotation $R_{\hat{n}}(\theta) = e^{-i\theta \hat{n} \cdot \vec{\sigma}/2}$, and **its eigenvectors are the two antipodes on its axis** — the only points it does not move. Changing basis chooses which axis to resolve along; it does not move the point.

The same three gates in three coordinate systems

	in $\{ 0\rangle, 1\rangle\}$	in $\{ +\rangle, -\rangle\}$	in <i>its own</i> eigenbasis	as a rotation
X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\text{diag}(1, -1)$	half-turn about $\hat{x}$
Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$\text{diag}(1, -1)$	half-turn about $\hat{z}$
H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$\text{diag}(1, -1)$	half-turn about $\frac{\hat{x} + \hat{z}}{\sqrt{2}}$

Columns 1 vs 2. X and Z **swap descriptions** under a change of basis: a bit flip in one coordinate system is a phase flip in the other. That is exactly $X = HZH$, $Z = HXH$.

Column 3. In its own eigenbasis, **every one of them is the same matrix**. They are one operator seen from three angles; only the axis differs.

M1 at size two: CNOT

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ in } \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}, \quad \text{CNOT} = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & -1 \end{pmatrix} \text{ in } \{|0\pm\rangle, |1\pm\rangle\}$$

since $\text{CNOT}|1\rangle|-\rangle = |1\rangle X|-\rangle = -|1\rangle|-\rangle$ and the other three are fixed.

In the right basis, CNOT is a controlled *phase* — and a diagonal gate has no preferred control wire: $\text{diag}(1, 1, 1, e^{i\theta})|x\rangle|y\rangle = e^{ixy\theta}|x\rangle|y\rangle$ is symmetric in $x \leftrightarrow y$.

“Control” and “target” are naming conventions

$(H \otimes H) \text{CNOT} (H \otimes H)$ is the CNOT with control and target swapped. The arrow the diagram draws is not in the operator.

That is phase kickback, one section early.

M2: tensor products, and where phases live

$\mathcal{H}_A \otimes \mathcal{H}_B$ is spanned by the products $|i\rangle \otimes |j\rangle$, and $\otimes$ is **bilinear**:

$$\left(\sum_i \alpha_i |i\rangle\right) \otimes \left(\sum_j \beta_j |j\rangle\right) = \sum_{i,j} \alpha_i \beta_j |i\rangle |j\rangle, \quad \boxed{|a\rangle \otimes (\lambda |b\rangle) = (\lambda |a\rangle) \otimes |b\rangle = \lambda |a\rangle |b\rangle}$$

Why bilinearity? It is not a new law of physics

It is what the tensor product *is*, forced by consistency: superposing a subsystem must be the same as superposing the joint system. And it encodes that amplitudes of independent systems *multiply*, exactly as independent probabilities do.

The boxed identity — **a scalar has no address** — is the single algebraic fact behind phase kickback. No phase ever “travels” between registers; the three expressions are literally the same vector.

Entanglement, and the phase dichotomy

$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$ is a product iff the coefficient matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ has **rank one**, i.e. $ad - bc = 0$.

Number of terms proves nothing: $|+\rangle|+\rangle$ has four terms and is a product. The Bell state has $ad - bc = \frac{1}{2}$: the two qubits **have no individual states**, only the pair does.

Global vs relative phase.

$$e^{i\gamma}|\Psi\rangle \equiv |\Psi\rangle \quad (\text{unobservable})$$

$$\frac{1}{\sqrt{2}}(|0\rangle + e^{i\theta}|1\rangle) : p(\pm) = \frac{1 \pm \cos \theta}{2}$$

An eigenvalue relation $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ only ever produces *global* phases.

Apply it to one branch of a superposition and the same number becomes **relative** — hence **physical**. Producing that situation is exactly what a *controlled* gate does.

Aside: entanglement was an objection before it was a resource

- **1935** — Einstein, Podolsky, Rosen: quantum mechanics predicts correlations between separated systems strong enough that measuring one appears to fix the other instantly; they argue the theory must be *incomplete*. Einstein: *spukhafte Fernwirkung*, spooky action at a distance.
- **1935** — Schrödinger, replying, names it *Verschränkung* (entanglement) and calls it “the characteristic trait of quantum mechanics”.
- **1964** — John Bell turns philosophy into an experiment: any theory with pre-existing local properties obeys an inequality that quantum mechanics violates.
- **1972** — Stuart Freedman and John Clauser, Berkeley: first laboratory test, using photon pairs from a calcium cascade. The inequality is violated.
- **1981–82** — Alain Aspect’s group at Orsay sharpens it; the 1982 run switches the analyser settings *while the photons are in flight*.
- **2022** — Nobel Prize in Physics: Clauser, Aspect, Zeilinger. (Loophole-free tests: 2015.)

For us: entanglement is a settled, nonlocal feature of the world — and in this algorithm it is both the resource and the hazard.

Phase kickback

A controlled- U is a **joint** operator: $c-U = |0\rangle\langle 0| \otimes \mathbf{1} + |1\rangle\langle 1| \otimes U$.

Theorem

If $U|u\rangle = e^{2\pi i\varphi}|u\rangle$ then $c-U[(\alpha|0\rangle + \beta|1\rangle) \otimes |u\rangle] = (\alpha|0\rangle + \beta e^{2\pi i\varphi}|1\rangle) \otimes |u\rangle$.

Proof. Linearity gives $\alpha|0\rangle \otimes |u\rangle + \beta|1\rangle \otimes e^{2\pi i\varphi}|u\rangle$; slide the scalar across the tensor sign (M2); factor out $|u\rangle$. $\square$

Three lines, and they consist entirely of moving a scalar. **Nothing physically travelled.**

(i) The target does not move.

An eigenvector is a fixed point; the gate can only emit a number. The target is a *catalyst* — reusable.

(ii) The scalar lands on the control.

What would have been a global phase is now *relative* between control branches. Information flowed *backwards*.

(iii) The registers stay a product.

No entanglement was created; the work register can be ignored from here.

Why kickback is not a trick

Expand $U = \sum_j e^{2\pi i \varphi_j} |u_j\rangle\langle u_j|$ inside $c\text{-}U$:

$$[c\text{-}U]_{\{|c\rangle \otimes |u_j\rangle\}} = \text{diag}(1, \dots, 1, e^{2\pi i \varphi_1}, e^{2\pi i \varphi_2}, \dots)$$

In the basis $\{|0\rangle, |1\rangle\} \otimes \{\text{eigenvectors of } U\}$, controlled- U is diagonal. Restricted to $\text{span}\{|0\rangle|u\rangle, |1\rangle|u\rangle\}$ it is exactly the phase gate $\text{diag}(1, e^{2\pi i \varphi})$ on the control.

Kickback is a diagonal matrix doing the only thing diagonal matrices do — read in the right coordinates.

The smallest instance, which you have already seen

$\text{CNOT}|+\rangle|-\rangle = |-\rangle|-\rangle$: a *bit flip on the control*, produced by a gate that “acts on the target”. Magic in the computational basis; $\text{diag}(1, 1, 1, -1)$ in $\{|0\rangle, |1\rangle\} \otimes \{|\pm\rangle\}$.

Shor's $|u_s\rangle$ is this trick with a finer protractor.

If the target is not an eigenvector: entanglement

$$c\text{-}U\left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \sum_j c_j |u_j\rangle\right] = \frac{1}{\sqrt{2}} \sum_j c_j (|0\rangle + e^{2\pi i \varphi_j} |1\rangle) |u_j\rangle$$

Each eigencomponent kicks back its own eigenphase. If two populated φ_j differ, the control state depends on which component it is glued to — the state no longer factors.

Quantitatively, the control alone is $\rho = \frac{1}{2} \begin{pmatrix} 1 & z \\ z^* & 1 \end{pmatrix}$ with $z = \langle \psi | U \psi \rangle$, giving fringes of visibility $|z| \leq 1$, equality iff $|\psi\rangle$ is an eigenvector.

Four descriptions, one fact

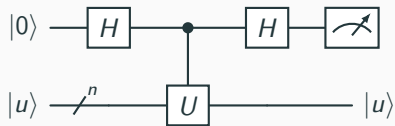
$|\psi\rangle$ is an eigenvector = $c\text{-}U$ creates no entanglement = the target records no which-path information = the fringes have full visibility.

(Careful: $c\text{-}U$ does *not* project the target onto an eigenstate. It is a fixed linear gate; the formula factorises only because $|u\rangle$ already *was* an eigenvector.)

One gate yields one phase, not one bit

After kickback the control is $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\varphi}|1\rangle)$ — whose computational statistics are 50:50 **for every φ** .

The phase is stored where a computational measurement cannot see it. It must be followed by a basis change — one H for one qubit, $F^\dagger$ for many — and **the design of that basis change is the design of the algorithm**.



$p(0) = \cos^2(\pi\varphi)$: a Mach-Zehnder interferometer. First H splits the paths, $c-U$ is a phase plate in one arm, second H recombines them.

The control cannot be dispensed with: uncontrolled U 's on $|u\rangle$ give only a global phase.

The one theorem the rest of the talk uses

Shifts are diagonal in the Fourier basis

Let $A_c|y\rangle = |(y+c) \bmod M\rangle$. The Fourier states $|\hat{\ell}\rangle = M^{-1/2} \sum_y e^{-2\pi i \ell y / M} |y\rangle$ satisfy

$$A_c|\hat{\ell}\rangle = e^{2\pi i c \ell / M} |\hat{\ell}\rangle \quad \Longleftrightarrow \quad F_M^\dagger A_c F_M \text{ is diagonal.}$$

(Substitute $y' = y + c$; the sum re-indexes cyclically.)

We will use it at four sizes:

- $M = 2$: $A_1 = X$, $F_2 = H$, $|\hat{1}\rangle = |-\rangle$ — the CNOT/ $|-\rangle$ example above. *done*
- $M = r$: U_a is a cyclic shift along an orbit of length r . *next*
- $M = 2^n$: run backwards, it *builds* an adder out of phase gates. *Act III*
- $M = 2^t$: the final $F^\dagger$ that reads the answer. *Act III*

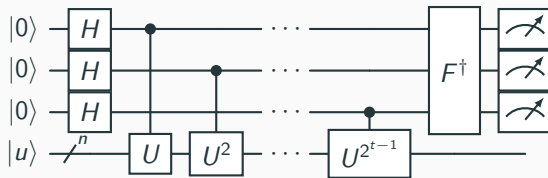
From one kickback to phase estimation

Put t control qubits in uniform superposition, $H^{\otimes t}|0\rangle^{\otimes t} = 2^{-t/2} \sum_x |x\rangle$ (“all exponents at once”), then apply the ladder $\sum_x |x\rangle\langle x| \otimes U^x$:

$$\frac{1}{2^{t/2}} \sum_x |x\rangle \otimes |u\rangle \mapsto \left[\frac{1}{2^{t/2}} \sum_{x=0}^{2^t-1} e^{2\pi i \varphi x} |x\rangle \right] \otimes |u\rangle$$

A phase ramp: as x increases by 1 the phase advances by the fixed angle $2\pi\varphi$ — a discrete plane wave of frequency φ , one bit of φ per control qubit.

Measuring the frequency of a wave is what Fourier transforms do.



Reading the ramp: a bank of matched filters

The ramp of frequency $\ell/2^t$ is literally $F|\ell\rangle$ — so $F^\dagger$ maps it to $|\ell\rangle$ exactly. In general the amplitude of outcome ℓ is

$$a_\ell = \frac{1}{2^t} \sum_{x=0}^{2^t-1} e^{2\pi i x \delta}, \quad \delta = \varphi - \frac{\ell}{2^t}.$$

Each ℓ is a hypothesis; $F^\dagger$ *unwinds* the ramp at that trial rate. Right hypothesis: 2^t unit vectors pointing the same way, adding coherently. Wrong: the arrows sweep the circle and the walk *curls up*.

Theorem

For ℓ nearest $2^t\varphi$: $\Pr[\ell] \geq 4/\pi^2 \approx 0.405$; the two nearest carry $\geq 8/\pi^2 \approx 0.81$.

The distribution is a Fejér kernel — the intensity pattern of a 2^t -slit diffraction grating. **Warning:** $4/\pi^2 < \frac{1}{2}$, so ordinary majority voting over repetitions does *not* amplify it; repetition instead raises the chance that *at least one* sample is usable.

Careful: this is only cheap for special U

The rungs are $U, U^2, U^4, \dots, U^{2^{t-1}}$. Building them by repetition costs

$$1 + 2 + 4 + \dots + 2^{t-1} = 2^t - 1$$

applications of U — exponentially many.

Phase estimation is efficient only for unitaries whose powers are cheap

For a generic black-box U , or a Hamiltonian evolution e^{iHt} , there is no shortcut and QPE offers *no speedup at all*.

Shor's algorithm exists because modular arithmetic hands us *both*:

- cheap powers (next slide), and
- a cheap circuit for one power (Act III).

Neither implies the other.

Why orders factor integers

Fix odd composite N (not a prime power), pick a with $\gcd(a, N) = 1$, let $r = \text{ord}_N(a) = \min\{k > 0 : a^k \equiv 1\}$.

Theorem

If r is even and $a^{r/2} \not\equiv -1 \pmod{N}$, then $\gcd(a^{r/2} - 1, N)$ is a nontrivial factor.

Proof. $x = a^{r/2}$ satisfies $N \mid x^2 - 1 = (x - 1)(x + 1)$, while $N \nmid x - 1$ (minimality of r) and $N \nmid x + 1$ (hypothesis). So the prime factors of N split between the two. $\square$

Fact. For N with $m \geq 2$ distinct prime factors and random a , the hypotheses hold with probability $\geq 1 - 2^{-(m-1)} \geq \frac{1}{2}$.

Everything now rests on computing r — for which no efficient classical algorithm is known.

From the integer a to a unitary

$$U_a|y\rangle = \begin{cases} |ay \bmod N\rangle, & 0 \leq y < N, \\ |y\rangle, & N \leq y < 2^n. \end{cases}$$

Since $\gcd(a, N) = 1$, $y \mapsto ay$ is a bijection — so U_a is a **permutation matrix**, automatically unitary. *Unitarity here is a number-theoretic condition*, and $\gcd(a, N) = 1$ does two jobs: it makes U_a a permutation, and it makes $\bar{a} = a^{-1}$ exist (needed in Act III).

The gift: the powers collapse

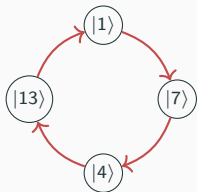
$$U_a^{2^j} = U_{a^{2^j} \bmod N}$$

A high power of U_a is **not a long circuit** — it is the *same* circuit with a different classical constant, precomputed by repeated squaring. This turns $2^t - 1$ applications into t rungs.

The cycle, and where the period hides

On the orbit $\{1, a, a^2, \dots, a^{r-1}\}$, U_a is a **cyclic shift of length exactly r** . By the shift theorem, its eigenvectors are the Fourier vectors of that cycle:

$$|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k / r} |a^k \bmod N\rangle, \quad U_a |u_s\rangle = e^{2\pi i s / r} |u_s\rangle.$$



computational basis: everything moves



eigenbasis: each axis spins in place

The order r is invisible in the matrix entries of U_a — all 0s and 1s — but written plainly in the eigenvalues. Phase estimation is the instrument that reads eigenvalues.

The circularity, and the trick that dissolves it

Writing down $|u_s\rangle$ requires knowing r — the very thing we want.

Lemma

$$\frac{1}{\sqrt{r}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$$

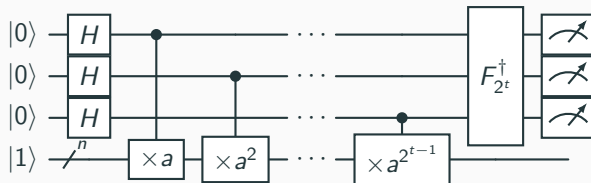
Proof. Exchange the sums: the coefficient of $|a^k\rangle$ is $\frac{1}{r} \sum_s e^{-2\pi i s k / r}$, a geometric series equal to $r\delta_{k,0}$. $\square$

$|1\rangle$ costs **one X gate** to prepare, and it is secretly the uniform superposition of all r eigenvectors — the Fourier transform of a delta function is flat, transported to the orbit.

Linearity runs the circuit on every eigenbranch at once; the branches are orthogonal (Lemma 2) and measurement conditions onto one.

The price: we get s/r for a *uniformly random* s , not chosen by us.

One circuit, two readings



A. Computational basis — it tabulates f

B. Eigenbasis — it kicks phases

$$\frac{1}{\sqrt{2^t}} \sum_x |x\rangle |a^x \bmod N\rangle$$

$$\frac{1}{\sqrt{r}} \sum_s \left[\frac{1}{\sqrt{2^t}} \sum_x e^{2\pi i x s / r} |x\rangle \right] \otimes |u_s\rangle$$

A lookup table of $f(x) = a^x \bmod N$:
argument in the top register, value in the
bottom, all 2^t rows at once. *Shor 1994.*

One phase ramp per eigenbranch, tagged by
 $|u_s\rangle$. *Kitaev;*
Cleve–Ekert–Macchiavello–Mosca.

These are the same vector. Substitute $|1\rangle = r^{-1/2} \sum_s |u_s\rangle$ and swap the sums.

No gate ever writes a phase

Every gate in that ladder is a **permutation matrix**. Below the final $F^\dagger$ there is no complex number anywhere in the circuit.

The phases in reading B are not added by anything. They are **what a permutation looks like in its own Fourier eigenbasis** — and they become physically meaningful only because the permutation was applied to a *superposition*, where a shuffle of basis labels turns into relative phases between branches.

Reading A: “build a periodic table and Fourier-sample it.”

Reading B: “estimate an eigenphase of U_a .”

One circuit, two descriptions — not two algorithms. The change of basis *is* the algorithm.

(Aside: that state is a Schmidt decomposition with r equal coefficients — the entanglement rank *is* the period.)

Why one run yields one eigenphase

$$\text{After } F^\dagger: \frac{1}{\sqrt{r}} \sum_s |\widetilde{2^t s/r}\rangle \otimes |u_s\rangle.$$

The tags $|u_s\rangle$ are **orthogonal** (Lemma 2), so the s -branches never interfere: for the counting register the state behaves as a classical mixture. Measurement is conditioning: outcome $\ell \approx 2^t s/r$ with s uniform, and the work register is projected onto that branch.

Contrast. Fed a single $|u_s\rangle$, the joint state stays a *product* throughout and the readout is deterministic for that one s . Fed $|1\rangle$, it is entangled — and **that is the feature**: the entanglement is what correlates each eigenvector with its own phase estimate.

A measurement that need not happen

Imagine measuring the work register in the eigenbasis *before* the circuit. That measurement commutes with every rung, so it can be slid to the end, where it cannot affect the counting statistics — hence it need never be performed. **The work register is never measured and never uncomputed.**

M3: the register you are not measuring casts its vote

Let the state before the final basis change be $|\Psi\rangle = \sum_x \alpha_x |x\rangle_A |g_x\rangle_B$ and apply $V (= F^\dagger)$ to A alone:

$$P(y) = \left\| \sum_x V_{yx} \alpha_x |g_x\rangle \right\|^2 = \sum_{x,x'} V_{yx} \overline{V_{yx'}} \alpha_x \overline{\alpha_{x'}} \langle g_{x'} | g_x \rangle$$

No branch-dependent record.

$\langle g_{x'} | g_x \rangle = 1$: the sum sits *inside* the modulus. Full interference.

A perfect record. $\langle g_{x'} | g_x \rangle = \delta_{xx'}$: the sum moves *outside* the modulus. Probabilities add where amplitudes should have.

Smallest instance. $|0\rangle \rightarrow H \rightarrow \text{CNOT onto a helper} \rightarrow H$:

$$\text{clean: } P(0) = \left| \frac{1}{2} + \frac{1}{2} \right|^2 = 1 \quad \text{copied: } P(0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Certainty becomes a coin flip — and nothing was measured. This is the double slit with a which-path detector: the fringes vanish even if nobody ever reads the detector.

Garbage in Shor's circuit

Reversible arithmetic built naively computes $|x\rangle|1\rangle|0\cdots 0\rangle \mapsto |x\rangle|a^x\rangle|g_x\rangle$, where g_x collects carries, comparison bits, partial products — and *depends on* x . Then

$$P(\ell) \propto \sum_{x,x'} e^{2\pi i(x-x')\ell/2^t} \langle a^{x'} | a^x \rangle \langle g_{x'} | g_x \rangle$$

The middle factor is the one we **want**: it confines interference to $x \equiv x' \pmod{r}$.
That is period finding.

The last factor is the **tax**: distinct carries $\Rightarrow$ orthogonal records $\Rightarrow$ every cross term dies $\Rightarrow P(\ell) = 2^{-t}$ exactly. **Uniform noise.**

The circuit is still perfectly unitary. **It simply computes nothing.**

So the question is never “is there a record?” but “**does the record separate branches I still need to add?**”

Uncomputation

Erasing scratch is many-to-one — not unitary. Discarding it changes nothing — the record still exists. The only move left is the one reversibility provides: **run the writing steps backwards**.

Theorem (Bennett): compute – copy – uncompute

Given a reversible circuit $C : |x\rangle|0\rangle \mapsto |x\rangle|f(x), g_x\rangle$ of size T , there is a circuit of size $\approx 2T$ realising

$$|x\rangle|y\rangle|0\rangle_{\text{scr}} \mapsto |x\rangle|y \oplus f(x)\rangle|0\rangle_{\text{scr}}$$

with every ancilla back to $|0\rangle$ in every branch.

Proof. Run C ; copy the answer with CNOTs; run $C^\dagger$. □

In place: $|x\rangle \mapsto |f(x)\rangle$ is impossible in general — *unless f is efficiently invertible*. Shor's multiplier uses $\bar{a} = a^{-1} \bmod N$ (extended Euclid): the second job of $\gcd(a, N) = 1$.

Roughly half of every circuit in Act III exists to run the other half backwards. That is not waste; it is the price of the interference.

III. Building it

What phase estimation demands of the circuit

Demand	Why	Discharged by
Unitary	no unitarity, no eigenphases	$\gcd(a, N) = 1 \Rightarrow$ permutation
Cheap powers	else the ladder costs $2^t - 1$ applications	$U_a^{2^j} = U_{a^{2^j \bmod N}}$
Controllable	uncontrolled U_a emits only a global phase	control threaded through every level
Garbage-free	scratch kills the interference	uncomputation at every level

A tower: the ladder is made of controlled multipliers, multipliers of modular adders, modular adders of plain adders, adders of Toffolis.

level	object	made of	multiplier
5	the ladder	$t \approx 2n$ rungs	$\times 2n$
4	c- $U_{a^{2^j}}$	$2 \times \text{CMULT} + n$ Fredkins	
3	CMULT	n doubly-controlled modular adders	$\times n$
2	modular ADD	5 constant additions + flag logic	$\times 5$
1	ADD	MAJ chain, CNOT, UMA chain	$\times 2n$ Toffoli
0	Toffoli	Clifford + T	$7T$; $4T$ if uncomputed

Level 0: the reversible gate set

gate	action	drawn as	Qiskit
CNOT = c - X	$ c, t\rangle \mapsto c, t \oplus c\rangle$	dot, $\oplus$	<code>qc.cx</code>
TOF (Toffoli) = cc - X	$ c_1, c_2, t\rangle \mapsto c_1, c_2, t \oplus c_1 c_2\rangle$	two dots, $\oplus$	<code>qc.ccx</code>
FRED (Fredkin) = c -SWAP	$ c, x, y\rangle \mapsto c, y, x\rangle$ iff $c = 1$	dot, $\times-\times$	<code>qc.cswap</code>
SWAP	$ x, y\rangle \mapsto y, x\rangle$ (three CNOTs)	$\times-\times$	<code>qc.swap</code>

Three facts to carry:

- The **Toffoli is a reversible AND**, and the only nonlinear gate here. Every AND costs one.
- **CNOTs implement exactly the $\mathbb{F}_2$ -linear maps** and are cheap — which is why counts are quoted in Toffolis ($\approx$ the T -count, which is what magic-state distillation bills).
- c - $P(\theta)$ is control/target symmetric — drawn as two dots joined by a line, no arrow.

Addition, the Fourier way (Draper)

The shift theorem at $M = 2^n$, **run backwards as a design principle**: adding a constant is a shift, and shifts are diagonal in the Fourier basis — so transform, apply phases, transform back.

$$|y\rangle \xrightarrow{\text{not}} \boxed{F} \text{---} \boxed{P(2\pi X 2^{i-n}) \text{ on each qubit } i} \text{---} \boxed{F^\dagger} \text{---} |y + X \bmod 2^n\rangle$$

n *uncontrolled single-qubit phase gates* do the whole addition. The composite $F^\dagger(\text{phases})F$ is again a 0/1 permutation — the complex numbers live only between the transforms.

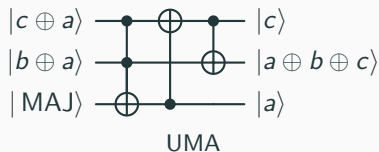
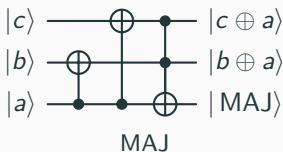
Why we will not use it

The smallest rotation is $2\pi/2^n$ — at $n = 2048$, about 10^{-616} radians. And fault-tolerantly, fine rotations are not Clifford+ T -exact, while Toffoli and CNOT are. **Hence: ripple-carry.**

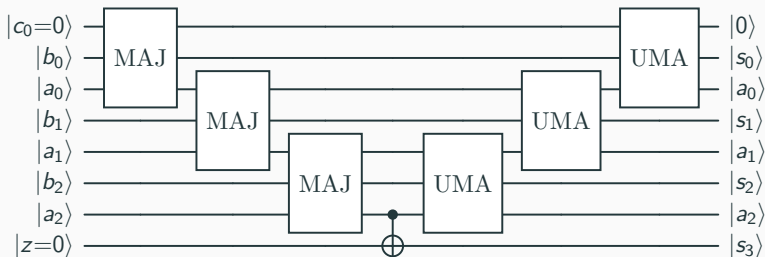
Addition, the ripple way (CDKM)

With carry-in $c_0 = 0$: $s_i = a_i \oplus b_i \oplus c_i$, $c_{i+1} = \text{MAJ}(a_i, b_i, c_i) = a_i b_i \oplus a_i c_i \oplus b_i c_i$.

Read it as a **cost statement**: sum bits are $\mathbb{F}_2$ -linear (CNOTs, free); carries are quadratic (a Toffoli each); and they form a chain (depth $\Theta(n)$). *That one line prices Shor's algorithm.*



The CDKM adder: uncomputation that pays for itself



$2n$ Toffolis, $4n+1$ CNOTs, $2n+2$ qubits, addition in place, **zero garbage**.

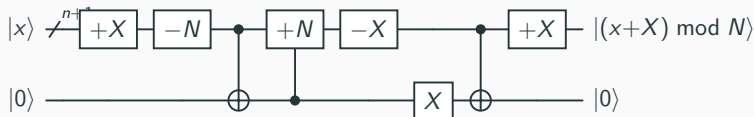
UMA erases the carry and writes the sum bit in the same three gates. Adding by hand you scribble carry digits above the columns; here you go back over the columns rubbing out each scribble, and the act of rubbing out deposits the answer.

Free corollaries: run it backwards to subtract; run the MAJ chain on $a, \bar{b}$ with carry-in 1 and the carry-out *is* the predicate $[a \geq b]$ — **comparison costs nothing extra**.

Modular addition: a quantum computer cannot do it

We want $x \mapsto (x + X) \bmod N$ — classically “add; if the result is $\geq N$, subtract N ”. But the register holds a superposition in which some branches overshot and some did not.

So the conditional becomes arithmetic: compute the condition *into a qubit*, use that qubit as a control — and then destroy the evidence.



Blocks 1–4 compute the answer. Blocks 5–7 exist entirely to destroy the evidence: since $x < N$, $\text{flag} = 1 \iff R \geq X$, so the flag is recomputable from the output.

You cannot remember whether you borrowed during a subtraction — but you can look at your answer, deduce that you must have, and then honestly forget.

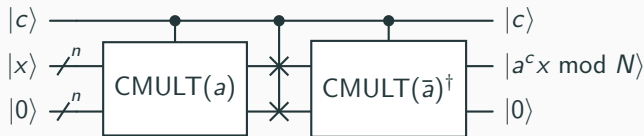
Multiplication, and the in-place trick

Shift-and-add: $a x \bmod N = \left(\sum_i x_i (2^i a \bmod N) \right) \bmod N$ — so the data bit x_i becomes a **control**. The data has turned into conditioning.

But CMULT writes into a *fresh* register, and the ladder must multiply the *same* register repeatedly. The leftover x must be *computed away*.

Theorem

With $\bar{a} = a^{-1} \bmod N$: $c\text{-}U_a = \text{CMULT}(\bar{a})^\dagger \cdot c\text{-SWAP} \cdot \text{CMULT}(a)$



$|x\rangle|0\rangle \rightarrow |x\rangle|ax\rangle \rightarrow |ax\rangle|x\rangle \rightarrow |ax\rangle|x - \bar{a}(ax)\rangle = |ax\rangle|0\rangle$. The dagger is just a minus sign on a classical number.

The bill

$$\underbrace{2n}_{\text{rungs}} \times \underbrace{2n}_{\text{adds}} \times \underbrace{5}_{\text{Toffoli}} \times \underbrace{2n}_{\text{Toffoli}} = \Theta(n^3) \text{ Toffoli} \quad (\approx 0.3n^3 \text{ with windowed arithmetic})$$

Three factors of n — rungs, multiplicand bits, addition bits — and nothing subtler is why the algorithm is cubic.

At $n = 2048$

Modular exponentiation: $\sim 10^9$ – 10^{10} Toffolis.

The (approximate) Fourier layer: $\sim 10^5$ gates.

The layer that carries all the conceptual content is about 0.003% of the circuit.

Whether RSA falls is the cost of multiplying two 2048-bit numbers, reversibly, in superposition.

Reading the answer classically

Measurement gives ℓ with $\left| \frac{\ell}{2^t} - \frac{s}{r} \right| \leq \frac{1}{2^{t+1}}$.

Theorem (Legendre)

If $|\xi - \frac{p}{q}| < \frac{1}{2q^2}$ with $\gcd(p, q) = 1$, then p/q is a convergent of the continued fraction of ξ .

Since $r < N$ we need $2^t \geq 2N^2$: **take** $t = 2n + 1$. The requirement comes from the *classical* post-processing — the quantum circuit does not care.

Then: expand $\ell/2^t$ by Euclid, take the eligible convergent, and **verify** $a^q \equiv 1$.

Failure modes, all cheap: ℓ off the peak ($\leq 1 - 4/\pi^2$); $s = 0$ (prob. $1/r$); $\gcd(s, r) = d > 1$ gives a divisor. A good s has probability $\varphi(r)/r$; two runs and an lcm recover r with constant probability.

The complete algorithm

1. Pick a at random; if $\gcd(a, N) > 1$, return it (lucky).
2. **Quantum:** run order finding — $t = 2n+1$ counting qubits, work register $|1\rangle$, ladder of $c-U_{a^{2^j}}, F^\dagger$, measure — to get ℓ .
3. Continued fractions on $\ell/2^t \rightarrow$ candidate r ; verify $a^r \equiv 1$, else re-run.
4. If r odd or $a^{r/2} \equiv -1$: back to step 1. Else return $\gcd(a^{r/2} - 1, N)$.

$O(n^3)$ gates, $O(n)$ qubits, constant expected repetitions — against $\exp(O(n^{1/3} \log^{2/3} n))$ classically.

$N = 15, a = 7, t = 11$

Orbit $1 \rightarrow 7 \rightarrow 4 \rightarrow 13 \rightarrow 1$, so $r = 4$. Outcomes $\ell \in \{0, 512, 1024, 1536\}$, each $\frac{1}{4}$. $\ell = 1536$: $\frac{1536}{2048} = \frac{3}{4} = [0; 1, 3] \Rightarrow r = 4$; $7^2 = 49 \equiv 4 \not\equiv -1$; $\gcd(3, 15) = 3$, $\gcd(5, 15) = 5$. **$15 = 3 \times 5$.**

Caveat: $(\mathbb{Z}/15)^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/4$, so r is always a power of two, $r \mid 2^t$, phase estimation is exact — and the continued-fraction step is never exercised. Every hardware demo uses $N = 15$ anyway. The honest test is $N = 21$.

Where the morals landed

The same fact did the work throughout — **shifts are diagonal in the Fourier basis**: at $M = 2$ it was $X = HZH$ and the CNOT/ $|-\rangle$ kickback; at $M = r$ it gave U_a 's spectrum; at $M = 2^n$, run backwards, it built the Draper adder; at $M = 2^t$ the final $F^\dagger$ read the answer.

M1 — the ladder is a permutation in one basis and a phase generator in another, and the change of basis *is* the algorithm.

M2 — every phase was moved by sliding a scalar across a tensor sign; only phases relative between branches were ever measurable.

M3 — the whole machine is one long fight to keep branches indistinguishable — uncomputation, flag re-derivation, in-place multipliers — until the one moment interference is spent, after which measurement is mere conditioning.

Unitaries are rotations; eigenvectors are the axes. U_a 's axes write the answer as an *angle*, and $F^\dagger$ is the final rotation that turns that angle into a *position* you can read.

Why it is hard to build — the same statement, backwards

The output depends on relative phases across $2^t \geq N^2$ branches staying coherent through $\sim n^3$ Toffolis.

Any stray coupling to an environment, $\sum_x \psi_x |x\rangle |E_0\rangle \rightarrow \sum_x \psi_x |x\rangle |E_x\rangle$, is the master formula again — with the environment as **the register you failed to uncompute**. Every cross term is taxed by $\langle E_{x'} | E_x \rangle$, and complete decoherence flattens the output to exactly 2^{-t} .

Decoherence is not noise added on top of a good answer; it is the continuous erasure of the only feature the algorithm produces.

Needed error rate $\sim 10^{-11}$ per gate, against $\sim 10^{-3}$ physical — hence error correction, and Toffolis paid for in magic states.

The feature that makes Shor work and the feature that makes it hard to build are the same one. Fault tolerance is the price of using a phase as a data register.

tensor products → controlled operations → eigenvectors and
kickback

→ entanglement → which-path records → uncomputation
→ phase estimation → Shor

Backup: the $4/\pi^2$ bound

For $\delta \neq 0$, sum the geometric series and use $|1 - e^{i\theta}| = 2|\sin(\theta/2)|$:

$$|a_\ell| = \frac{1}{2^t} \cdot \frac{|\sin(\pi 2^t \delta)|}{|\sin(\pi \delta)|}$$

Bound the numerator below by the chord ($|\sin \pi \theta| \geq 2|\theta|$ for $|\theta| \leq \frac{1}{2}$, at $\theta = 2^t \delta$) and the denominator above by the arc ($|\sin u| \leq |u|$):

$$|a_\ell| \geq \frac{1}{2^t} \cdot \frac{2 \cdot 2^t |\delta|}{\pi |\delta|} = \frac{2}{\pi}.$$

Tight: at $|\delta| = \frac{1}{2^{t+1}}$ exactly, $\Pr[\ell] \rightarrow 4/\pi^2$ from above as t grows.

Backup: the code, in outline

```
def maj(qc, c, b, a):
    qc.cx(a, b); qc.cx(a, c); qc.ccx(c, b, a)

def uma(qc, c, b, a):
    qc.ccx(c, b, a); qc.cx(a, c); qc.cx(c, b)

def cdkm_add(qc, a, b, c0, z): #  $|a\rangle|b\rangle \rightarrow |a\rangle|(a+b) \bmod 2^n\rangle$ 
    triples = [(c0, b[0], a[0])] + [(a[i-1], b[i], a[i]) for i in range(1, len(a))]
    for t in triples: maj(qc, *t)
    qc.cx(a[-1], z)
    for t in reversed(triples): uma(qc, *t)

def c_ua(qc, c, x, b, flag, A, N): #  $|x\rangle|0\rangle \rightarrow |A^c x \bmod N\rangle|0\rangle$ 
    c_mult_acc(qc, c, x, b, flag, A, N)
    for i in range(len(x)): qc.cswap(c, x[i], b[i])
    c_mult_acc(qc, c, x, b, flag, (-pow(A, -1, N)) % N, N)

def order_circuit(A, N, t):
    qc.x(tgt[0]); qc.h(ctr)
    for j in range(t):
        c_ua(qc, ctr[j], tgt, anc[:n], anc[n], pow(A, 2**j, N), N)
    inverse_qft(qc, ctr); qc.measure(ctr, out)
```

Illustrative and deliberately unchecked: it documents the block $\leftrightarrow$ theorem correspondence, not a tested implementation.

Backup: is position ever the computational state?

Yes — in three different senses. Only the third is the regime of this talk.

1. Position and momentum are the degrees of freedom (infinite-dimensional)

Continuous-variable quantum computing. The unit is a *qumode*: a mode of the light field whose quadratures obey $[\hat{x}, \hat{p}] = i\hbar$ — formally the position and momentum of a harmonic oscillator. Gates are squeezing, displacement, beamsplitters, plus one non-Gaussian element. Universality proved by **Lloyd & Braunstein, Phys. Rev. Lett. 82, 1784 (1999)**; largest photonic demonstration to date, 216 squeezed modes: **Madsen et al., Nature 606, 75 (2022)** (Borealis).

2. One qubit encoded into position space

GKP codes — **Gottesman, Kitaev & Preskill, Phys. Rev. A 64, 012310 (2001)**, “Encoding a qubit in an oscillator”: the logical states are grids of spikes in x - p phase space, so a small shift in position is a correctable error. Realised in the *motion of a single trapped ion* (**Flühmann et al., Nature 566, 513 (2019)**, ETH Zürich) and in a superconducting cavity (**Campagne-Ibarcq et al., Nature (2020)**, Yale). The first error correction to beat break-even used this encoding: **Sivak et al., Nature (2023)**.

3. Position coarse-grained to two places

Dual-rail photonics: $|0\rangle$ = photon in waveguide A, $|1\rangle$ = waveguide B — which *place* the photon is in (**Knill, Laflamme & Milburn, Nature 409, 46 (2001)**). **Charge qubit:** which of two quantum dots holds one electron (**Petersson et al., Phys. Rev. Lett. 105, 246804 (2010)**) — coherent, but badly exposed to charge noise, which is why silicon dots now encode in *spin* instead.

4. Position as plumbing, not memory — **this talk**

Trapped ions keep the qubit in internal electronic levels and use the ions' shared *motional* mode as the bus that couples them (**Cirac & Zoller, Phys. Rev. Lett. 74, 4091 (1995)**); neutral-atom machines move atoms in tweezers to choose which pairs interact. The stored qubit is a finite internal property — a two-outcome wavefunction, exactly as on slide 3.